

TECHNOLOGIES FOR MOTOR BEHAVIOUR ANALYSIS AND VIRTUAL MODELLING [1]



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PRACTICE 3

IMUs

Laura Cruciani



INERTIAL MEASUREMENT UNITS



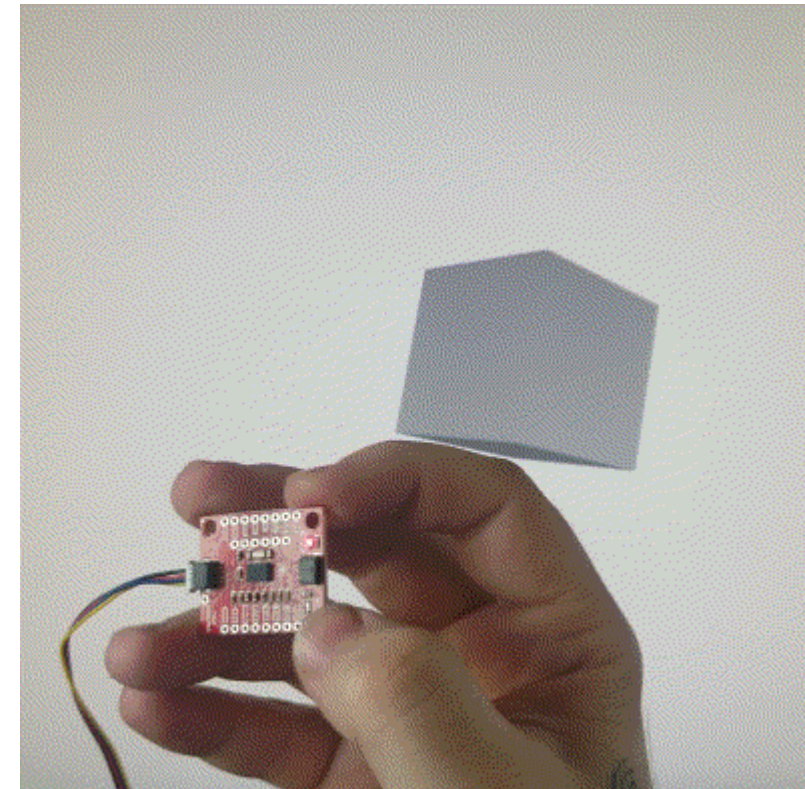
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They're basically composed by three main elements

- ▶ Accelerometer (Acceleration – 3DOF)
 → measures acceleration x, y, z
- ▶ Gyroscope (Angular Speed – 3DOF)
 → angular speed x, y, z
- ▶ Magnetometer (Magnetic Field – 3DOF)
 → magnetic field

9DOF fusion → Orientation

if we join them we obtain the orientation in space of our device

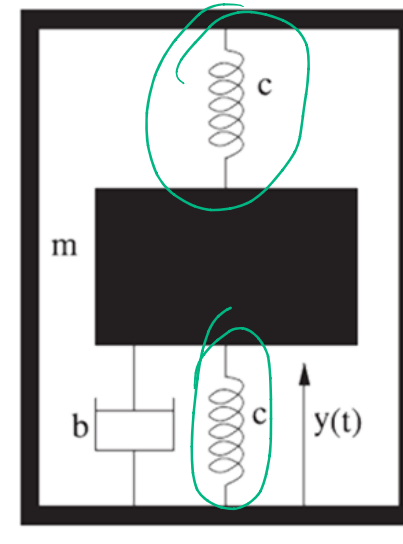
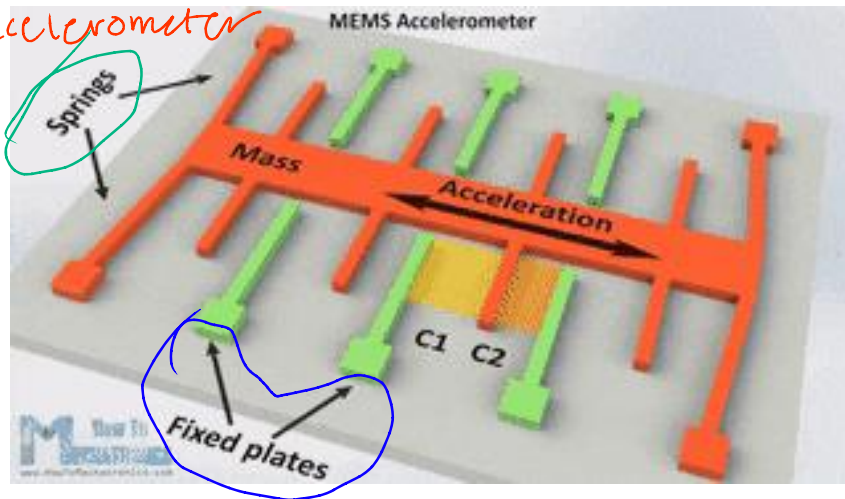


ACCELEROMETER: BASICS

ACCELEROMETER

An accelerometer is a tool that measures the **acceleration** (the rate of change of velocity). At rest relative to the Earth's surface will indicate approximately 1 g.

measures changing capacitance in capacitive accelerometer



$$\ddot{y}(t) + 2\xi\omega_n\dot{y}(t) + \omega_n^2 y(t) = \ddot{x}(t)$$

$$\omega_R = \omega_n \sqrt{1 - 2\xi^2}$$

$$\omega_n = \sqrt{\frac{c}{m}}$$

$$\xi = \frac{b}{2\sqrt{c \cdot m}}$$

natural freq

mass moves and changes value of capacitance

ACCELEROMETER: BASICS

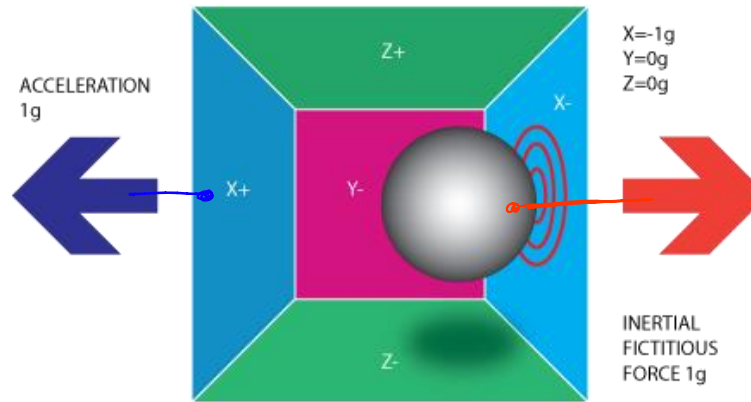
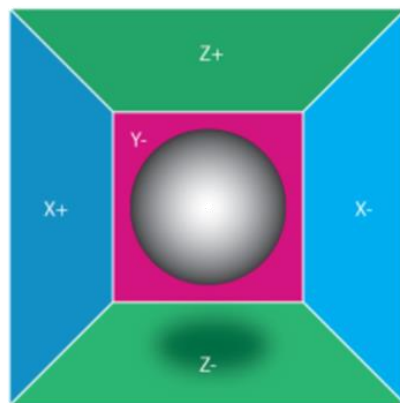


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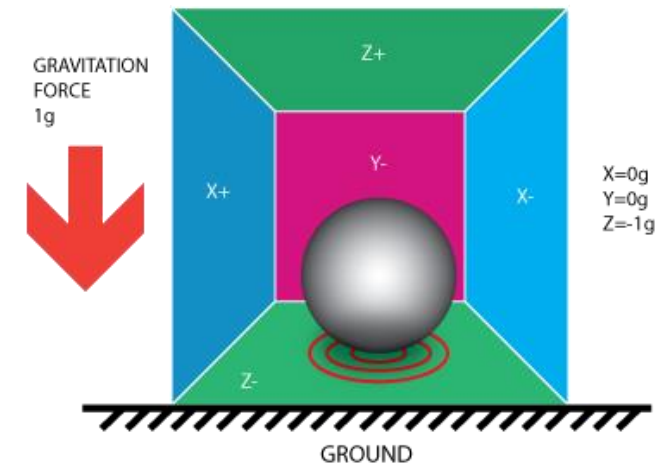
ACCELEROMETER Doesn't measure directly the acceleration but

An **acceleration** causes an **inertial force** that is captured by the accelerometer.

imagining wall of box is pressure sensitive



accelerometer measures force produced by acceleration



Even if no acceleration, gravity in earth produces a $-1g$

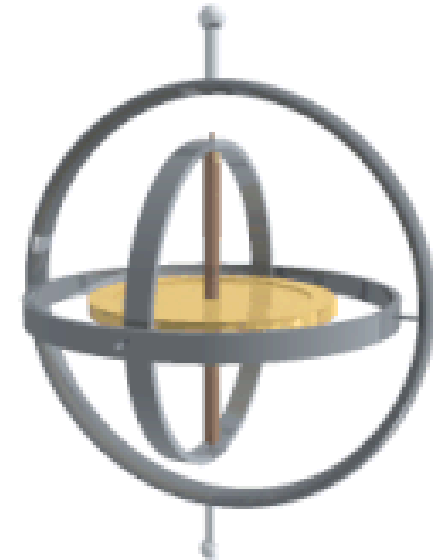
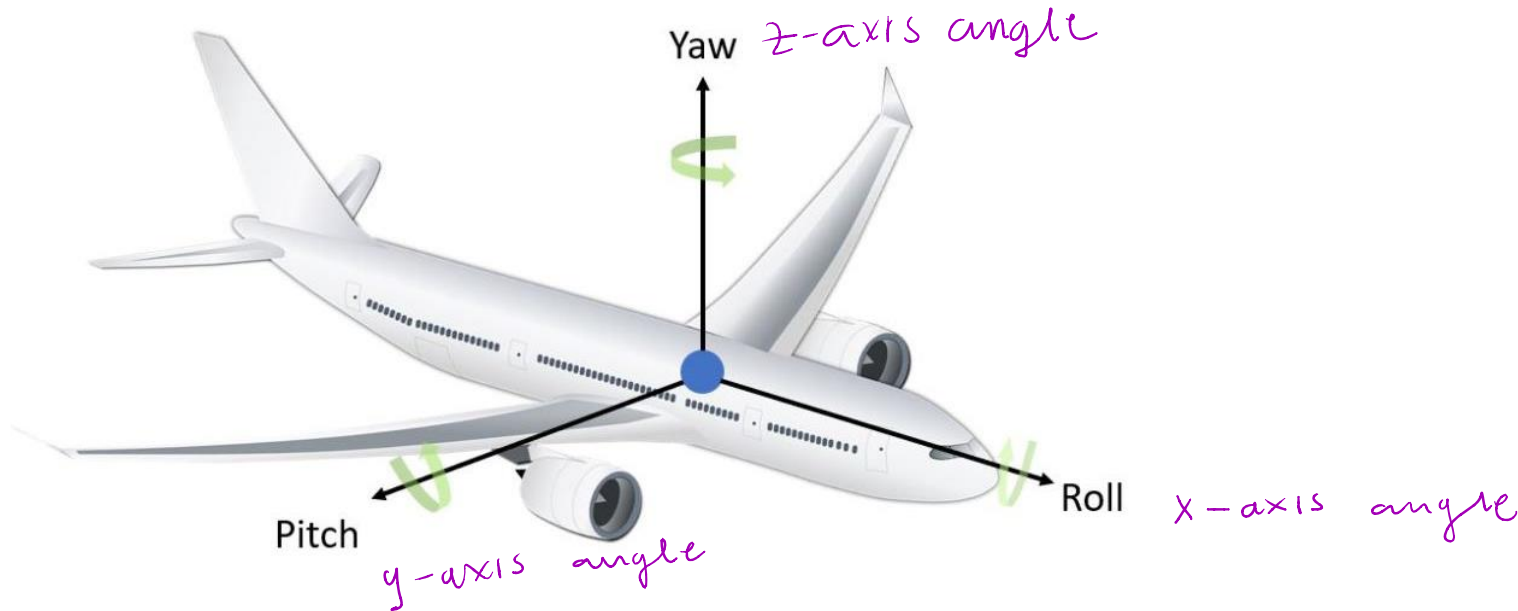
GYROSCOPE: BASICS



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GYROSCOPE

Gyroscopes measures **angular velocity** on three axes: **pitch**, **roll** and **yaw**.
Orientation: $\int \dot{w}$.



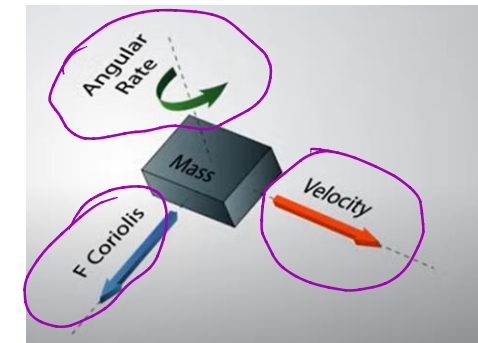
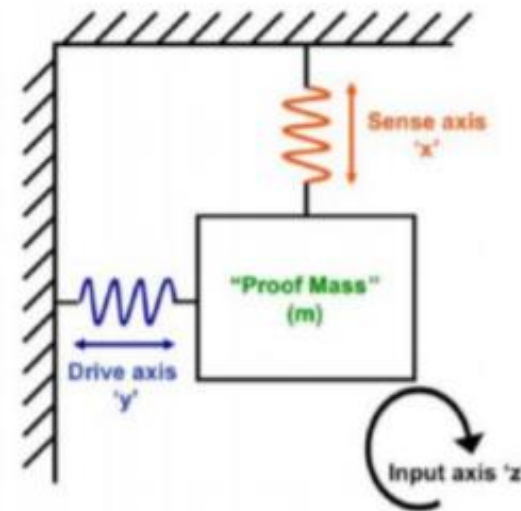
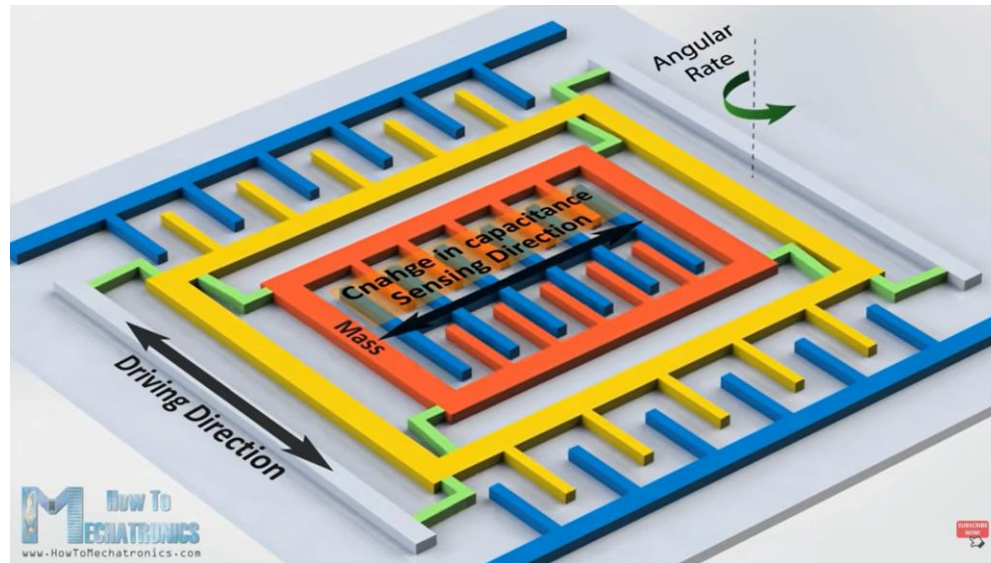
GYROSCOPE: BASICS



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GYROSCOPE Also in this case we have a mass
which experiences a certain velocity

Gyroscopes measures **angular velocity** on three axes: **pitch**, **roll** and **yaw**.
Orientation: $\int \dot{w}$.



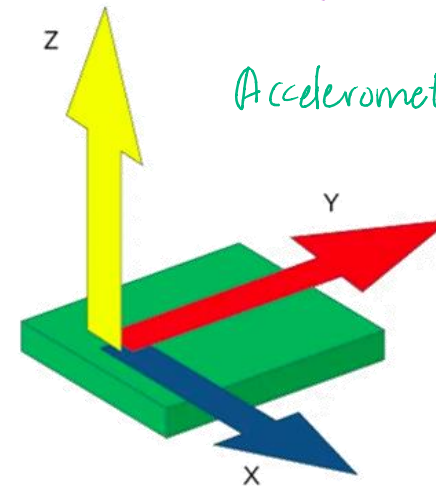
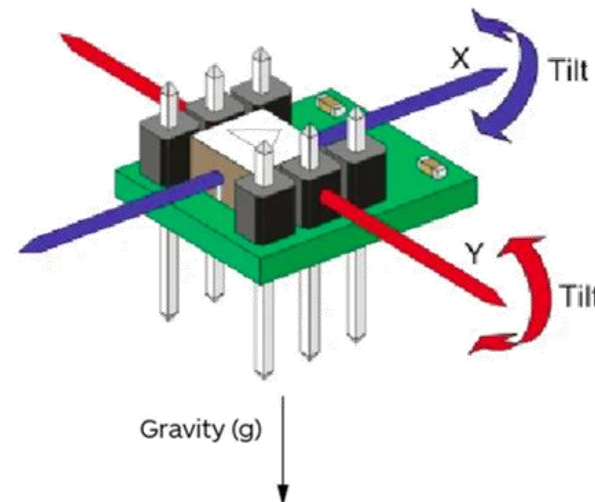
INERTIAL MEASUREMENT UNITS



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ACCELEROMETER + GYROSCOPE

Accelerometer data → Noisy (**jittering**) but it has the information with respect to the ground but **no yaw estimation**. → so we can't have orientation,



Accelerometer moving in around X or Y axis, we see no change on Z-axis, no change in the yaw-angle

INERTIAL MEASUREMENT UNITS



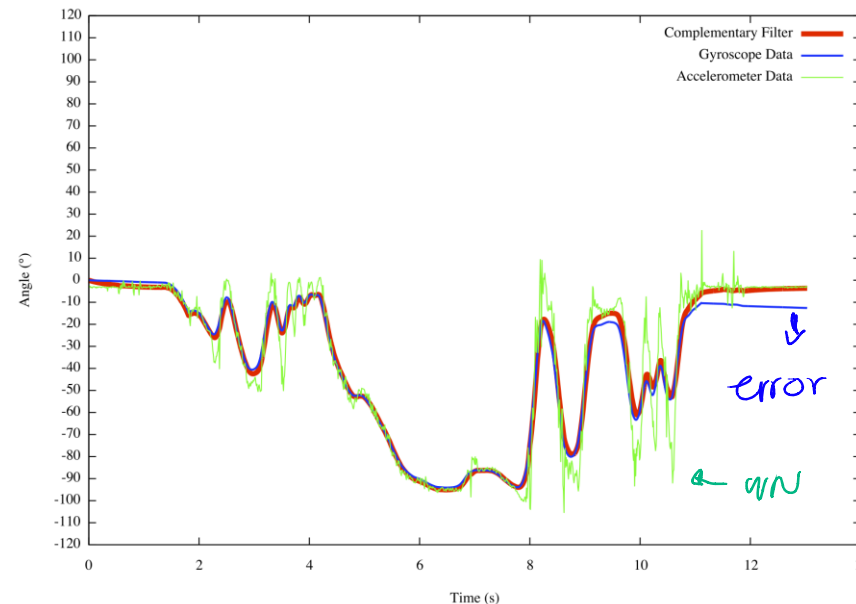
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ACCELEROMETER + GYROSCOPE

Accelerometer data → Noisy (**jittering**) but it has the information with respect to the ground.

Gyroscope data → **Drift** would occur eventually

we're integrating its value, at the end we have an error due to integration and this will sum-up in time, which increases



error increases, so gyroscope alone is not suitable

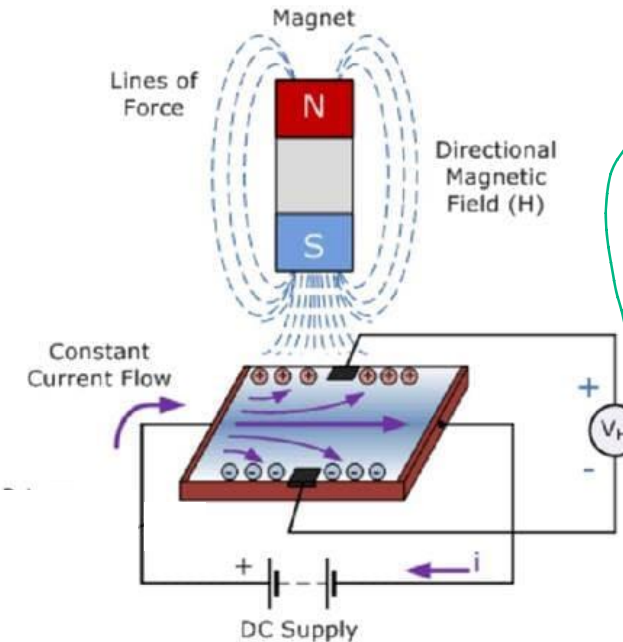
MAGNETOMETER: BASICS



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MAGNETOMETER To obtain the value of the yaw-angle, we need to integrate is the magnetometer

Magnetometer measures **Earth's magnetic field** by using the **Hall effect**.



V_H is $\perp$ to the magnetic field, and it depends on its magnitude and orientation

MAGNETOMETER: BASICS



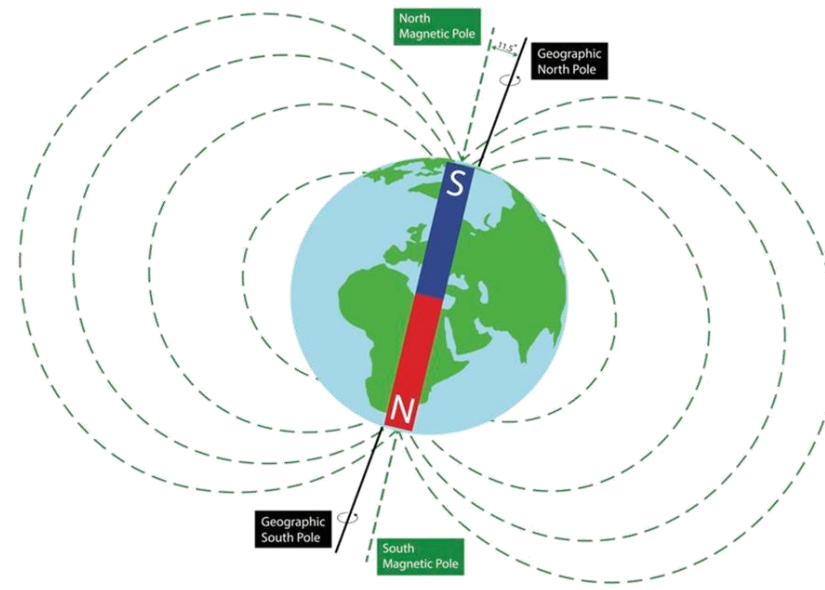
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MAGNETOMETER is like a compass that points the magnetic north

Geographic North: it is the fixed point where the **Earth's axis of rotation** meets its surface.

Magnetic North: it is the direction that a compass points to as it aligns with the **Earth's magnetic field**

is the truth one
and fixed one



MAGNETOMETER: BASICS

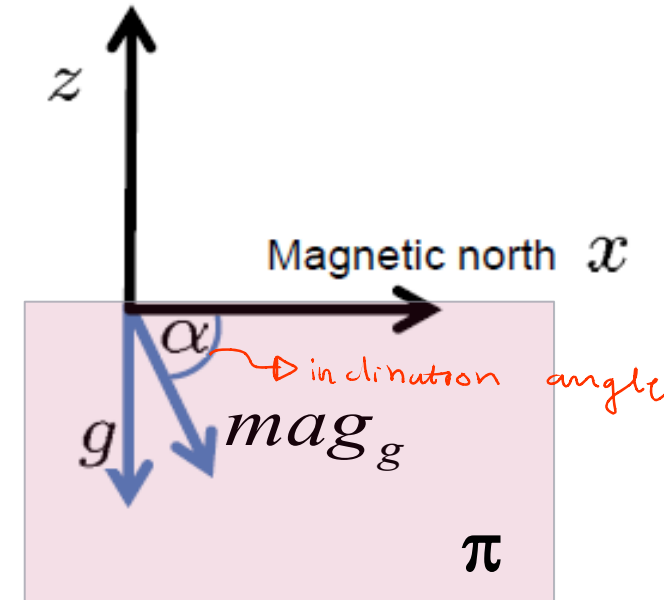
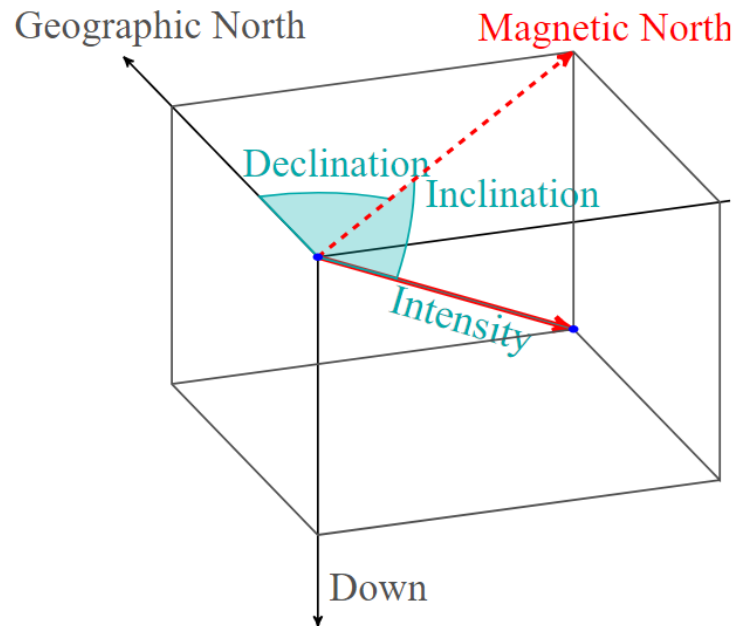


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MAGNETOMETER The difference between the geographic North and magnetic north is defined by the inclination angle

Declination: angle between Magnetic and Geographic North.

Inclination: angle between the Magnetic North and the direction in which the compass is pointing.

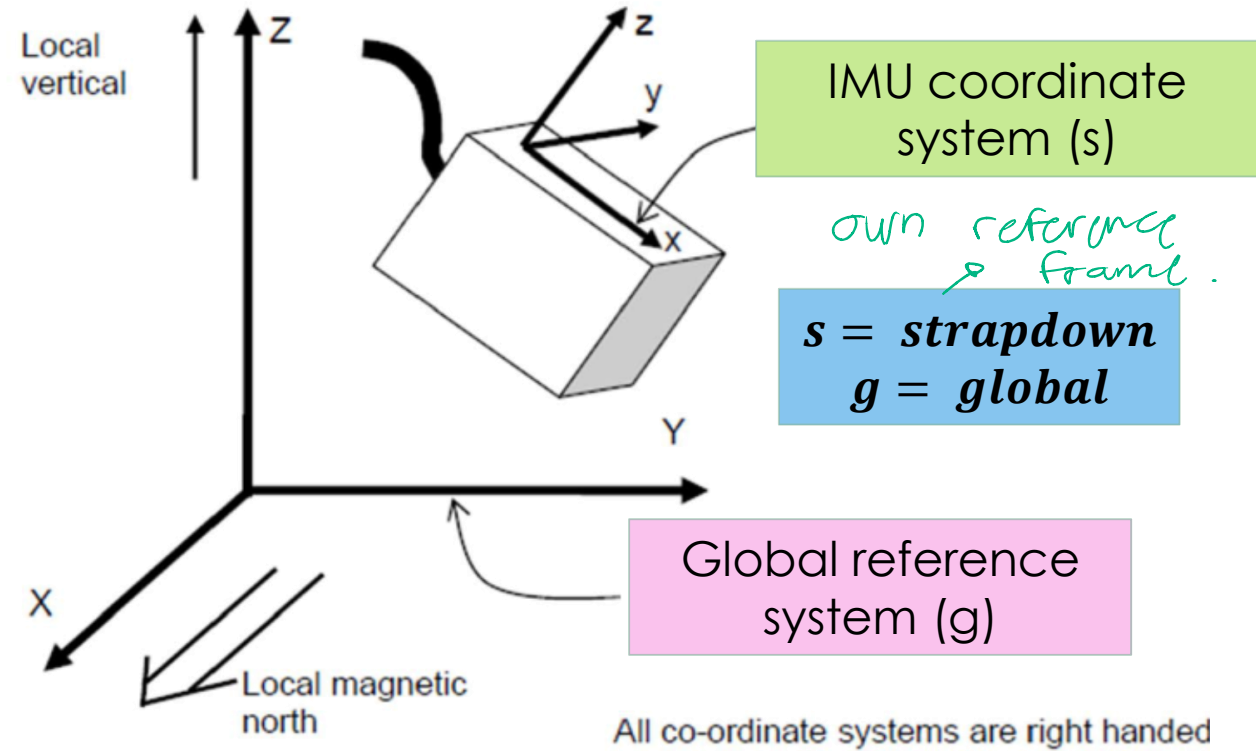


IMU NOTATION



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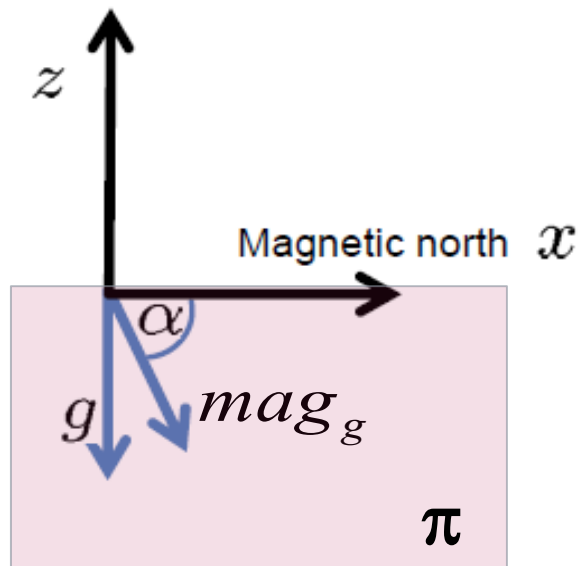
$\overset{s}{y}^{\omega} = \text{angular speed}$
 $\overset{s}{y}^a = \text{linear acceleration}$
 $\overset{s}{y}^m = \text{magnetic field}$



MAGNETOMETER REPRESENTATION



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$${}^g mag = \begin{bmatrix} h \cos \alpha \\ 0 \\ -h \sin \alpha \end{bmatrix}$$

Inclination angle

angle

magnitude

Field strength

Magnetometer reading

we can go from global reference to own reference by multiplying R

$${}^s y^m = {}^s mag = {}^s R_g {}^g mag$$

Rotation matrix

Paris:

- field inclination = 64°
- field strength = 48205 nT

ROTATION MATRIX PRINCIPLES



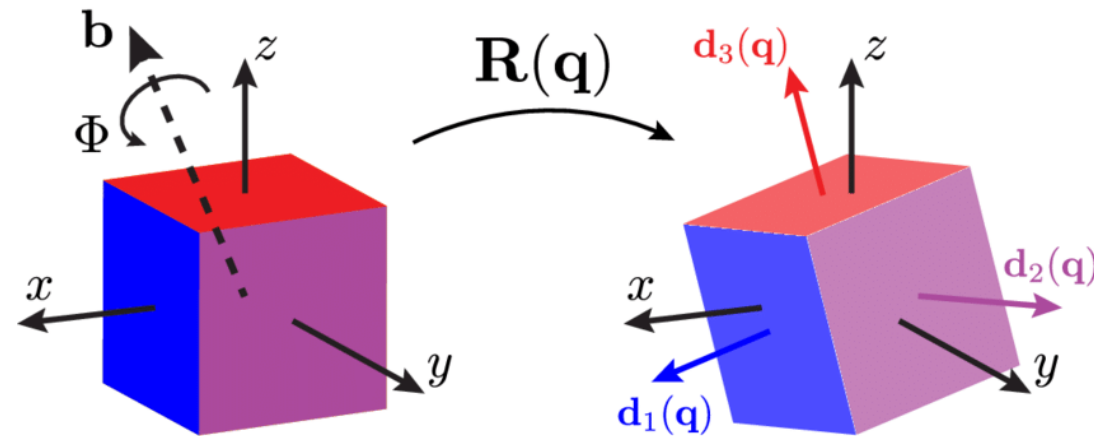
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Depending on along which axis we're rotating

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



And if we multiply them

$$R = R_z(\alpha) R_y(\beta) R_x(\gamma) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}^{\text{yaw}} \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}^{\text{pitch}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix}^{\text{roll}}$$

we obtain the final rotation

FINDING ROTATION MATRIX



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From the accelerometer: we obtain R_3 (z-axis)

$${}^sR_g = [r_1 \ r_2 \ r_3]_{3 \times 3}$$

$${}^s y^a = {}^s \ddot{p} + {}^s g = {}^s R_g (\cancel{{}^g \ddot{p}} + {}^g g) \approx {}^s R_g {}^g g$$

we obtain the rotation in
its own reference frame

$$\frac{{}^s y^a}{\|{}^s y^a\|} = + {}^s R_g \frac{{}^g g}{\|{}^g g\|}$$

$$\frac{{}^s y^a}{\|{}^s y^a\|} = + [r_1 \ r_2 \ r_3] \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} = -r_3 (\text{vector } 3 * 1) \longrightarrow$$

Direction z axis in global frame

FINDING ROTATION MATRIX



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From the accelerometer and magnetometer:

$${}^sR_g = [r_1 \ r_2 \ r_3]_{3 \times 3}$$

*we obtain r_2
by means of the cross product of*

$$\frac{{}^s y^m \times {}^s y^a}{\| {}^s y^m \times {}^s y^a \|} = \mathbf{r}_2$$

Direction y axis in global frame

we obtain the direction of the y-axis.

$$r_2 \times r_3 = r_1$$

Direction x axis in global frame

*And finally we obtain the final one.
with a simple cross-product.*

FINDING ROTATION MATRIX



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$${}^s y^a = {}^s \ddot{p} + {}^s g = {}^s R_g ({}^g \ddot{p} + {}^g g)$$

$${}^g \ddot{p} = {}^g R_s {}^s y^a - {}^g g \quad {}^g R_s = {}^s R_g^{-1}$$

then we get rid of the gravity

Body acceleration (without g)

IMU SENSOR

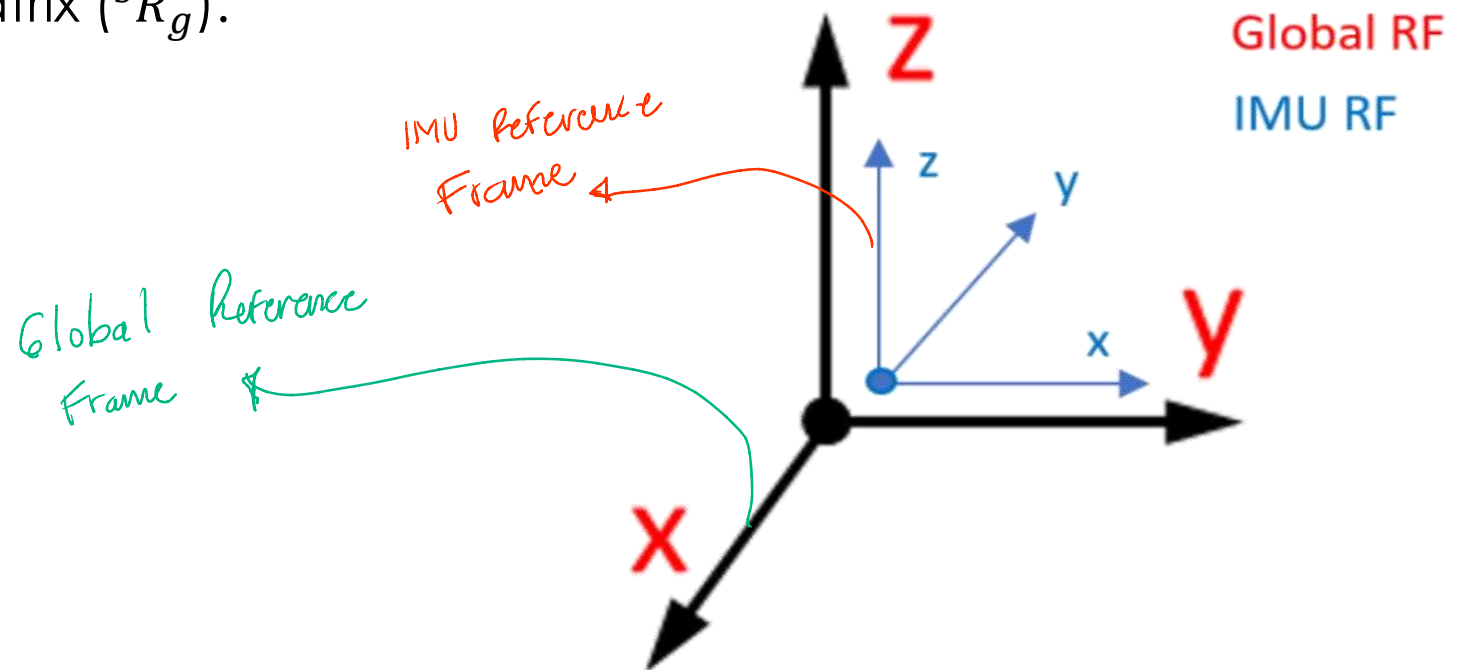


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EXERCISE 1

Consider the Global and IMU's reference frames in figure and select the right components of the rotation matrix (sR_g).

We need to find the
rotation matrix sR_g



IMU SENSOR



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SOLUTION

$${}^sR_g = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

	g_x	g_y	g_z
s_x	0	1	0
s_y	-1	0	0
s_z	0	0	1

This can be performed by just looking with no need of previous calculation

Global Reference Frame.

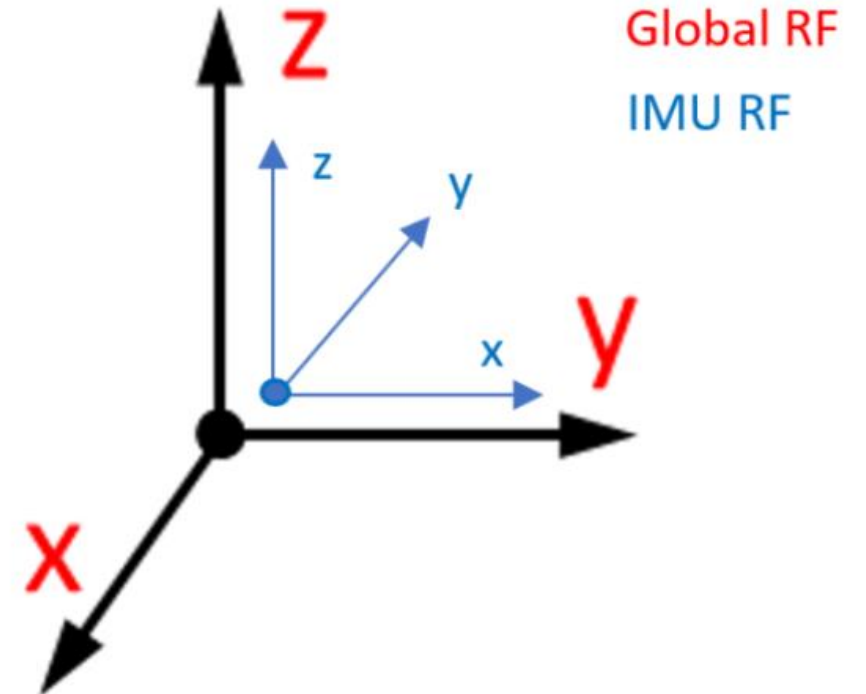
IMU
Ref
frame

	x	y	z
x	0	1	0
y	-1	0	0
z	0	0	1

Opposite

Same direction

same z axis



IMU SENSOR



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EXERCISE 2

A boxer's elbow acceleration is measured during a match by using IMU. Gravity is given with respect to the global reference frame (G):

$${}^G g = [0, 0, -9.81] \text{ m/s}^2$$

Here you see z -axis were aligned thus $\vec{r}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ can be inferred.

When the linear acceleration due to the elbow movement is negligible (steady position of arm), the measured linear acceleration in the IMU's reference frame (s) is:

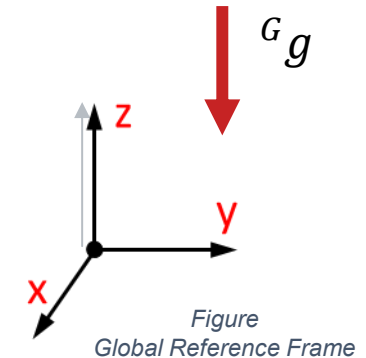
$${}^s y_{steady}^a = [0, 0, -9.81] \text{ m/s}^2$$

The measured magnetic field is:

$${}^s y^m = [0, 0.5, -0.60] * 10^{-4} \text{ T}$$

Measured linear acceleration during the experiments of elbow movement:

$${}^s y_{moving}^a = [-0.1, 0, -9.96] \text{ m/s}^2$$



IMU SENSOR



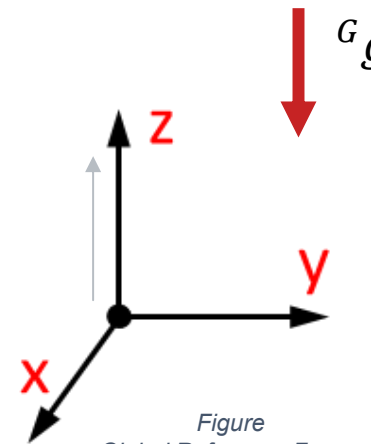
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REQUESTS

Compute linear accelerations that elbow is subjected in the global reference frame. Note that, magnetic north is aligned with the x-axis of the Global Frame

we want the acceleration from the global reference frame.

we need to compute the rotational matrix that



IMU SENSOR



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SOLUTION

in a steady case, it is equal to the gravitation vector in global reference plane

$$\frac{s y_{steady}^a}{\|s y_{steady}^a\|} = {}^s R_g \frac{{}^G g}{\|{}^G g\|} \rightarrow s y_{steady}^a = [r_1 \ r_2 \ r_3] \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \rightarrow s y_{steady}^a = -r_3 \rightarrow \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} = -r_3 \rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = r_3$$

Acceleration in steady state is $-r_3$ → then r_3 is equal

$$\frac{s y^m \times s y^a}{\|s y^m \times s y^a\|} = r_2$$

$$\rightarrow r_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

r_2 is found by cross-product of accelerometer x magnetometer over its norm

we obtain r_1 when having r_2 and r_3

$$r_1 = r_2 \times r_3 \rightarrow r_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$$

$$\mathbf{b} = b_1 \mathbf{i} + b_2 \mathbf{j} + b_3 \mathbf{k}$$

$$\begin{array}{l} +ia_2b_3 \\ +ab_2k \\ +b_1ja_3 \\ -b_1a_2k \\ -ib_2a_3 \\ -a_1jb_3 \end{array} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

IMU SENSOR



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SOLUTION

Once the three vectors are found we obtain the matrix by simply

The problem is asking to calculate acceleration in moving phase

$$r_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad r_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \quad r_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \rightarrow {}^sR_g = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$r_1 \quad r_2 \quad r_3$

$${}^gR_s = {}^sR_g^{-1}$$

acceleration reading is

$${}^s y^a = {}^s \ddot{p} + {}^s g = {}^s R_g ({}^g \ddot{p} + {}^g g)$$

$${}^g \ddot{p} = {}^g R_s {}^s y^a - {}^g g$$

once we get the rotation matrix we can obtain the speed of ${}^g \ddot{p}$

$${}^g \ddot{p} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.1 \\ 0 \\ -9.96 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ -9.8 \end{bmatrix} \quad \rightarrow {}^g \ddot{p} = \begin{bmatrix} 0 \\ 0.1 \\ -0.15 \end{bmatrix}$$

IMU SENSOR



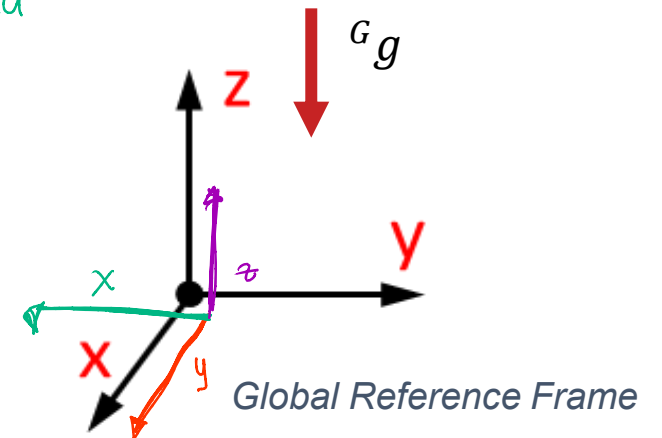
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EXERCISE 3

Consider the global frame given in the first question, draw the IMU reference frame with the global frame. Pay attention to the orientation with respect to Global Frame.

We have the global reference frame and we need to draw the IMU reference frame w.r.t of the global one

$$\text{Recall: } {}^gR_s = \begin{matrix} & \begin{matrix} s_x & s_y & s_z \end{matrix} \\ \begin{matrix} g_x \\ g_y \\ g_z \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$



IMU SENSOR

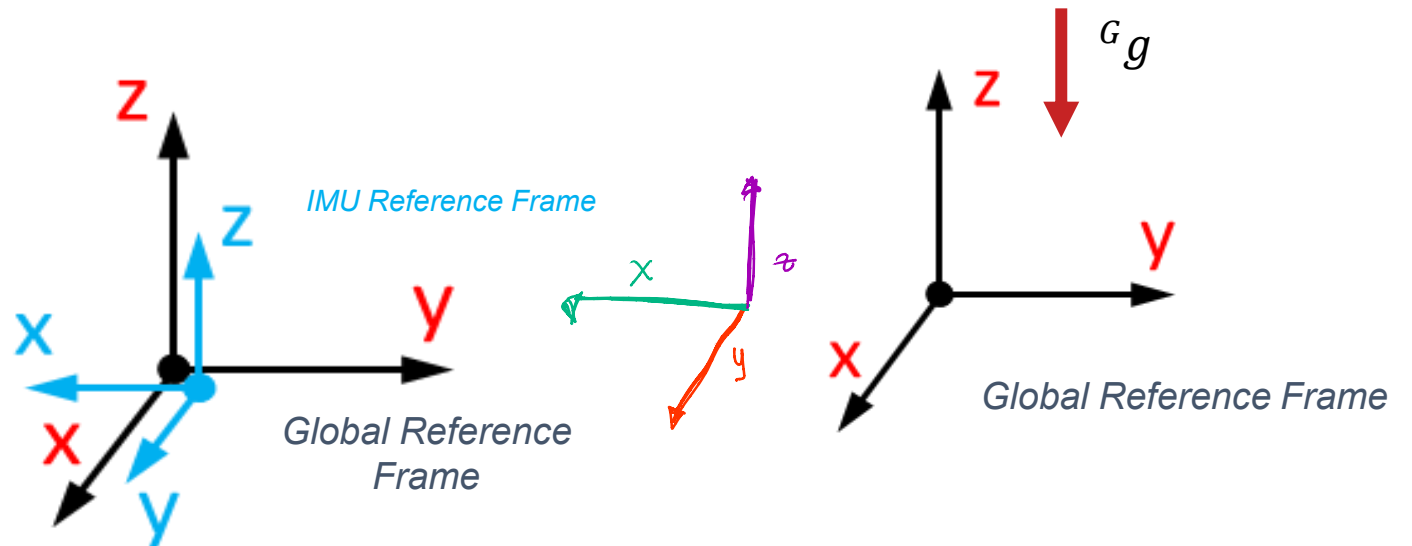


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EXERCISE 3

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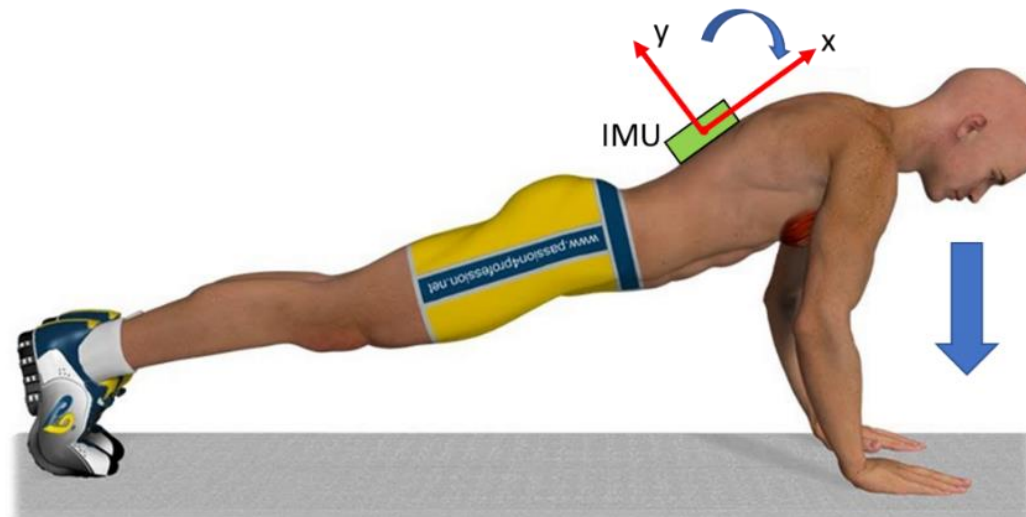
$$\text{Recall: } {}^gR_s = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



IMU SENSOR

EXERCISE 4

Consider the following scenario for the application of IMU, where IMU is used in order to measure the number of proper push-ups performed by the athlete. As seen in the figure, IMU (the green object) placed at the back of the athlete:



IMU SENSOR



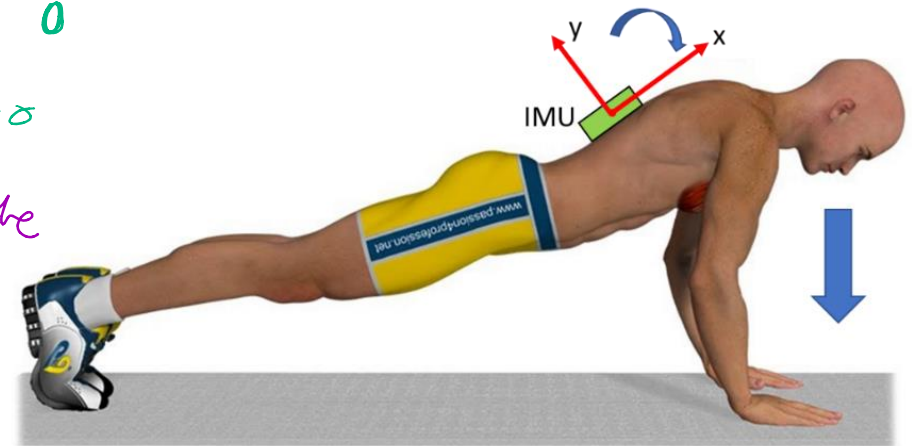
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EXERCISE 4

Right before athlete starting to lying close to the surface (body acceleration = 0), the IMU measures acceleration along its x-axis equal to -6.9367 m/s^2 . After that, the athlete starts lying close to the surface while IMU rotated in the given direction as a 15° .

Compute the acceleration component measured along the rotated IMU x and y-axis. (Gravity vector is 9.81 m/s^2 towards to ground)

Steady case body acceleration is equal to 0
and in x-axis acceleration is equal to
 -6.9367 m/s^2 then athlete does the
push-up and we get an
inclination angle of 15° w.r.t initial
one we want x-y acceleration after
push-up



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SOLUTION

$$\text{Initial Angle: } \cos^{-1}\left(\frac{-6.9367}{-9.81}\right) = 45^\circ$$

this is the initial angle is 45°

$$\text{Final Angle: } 45^\circ + 15^\circ = 60^\circ$$

After push-up, angle increases by 15°

$$IMU_x = -9.81 * \cos(\text{angle}_{\text{final}}) \frac{m}{s^2} = -4.9 m/s^2$$

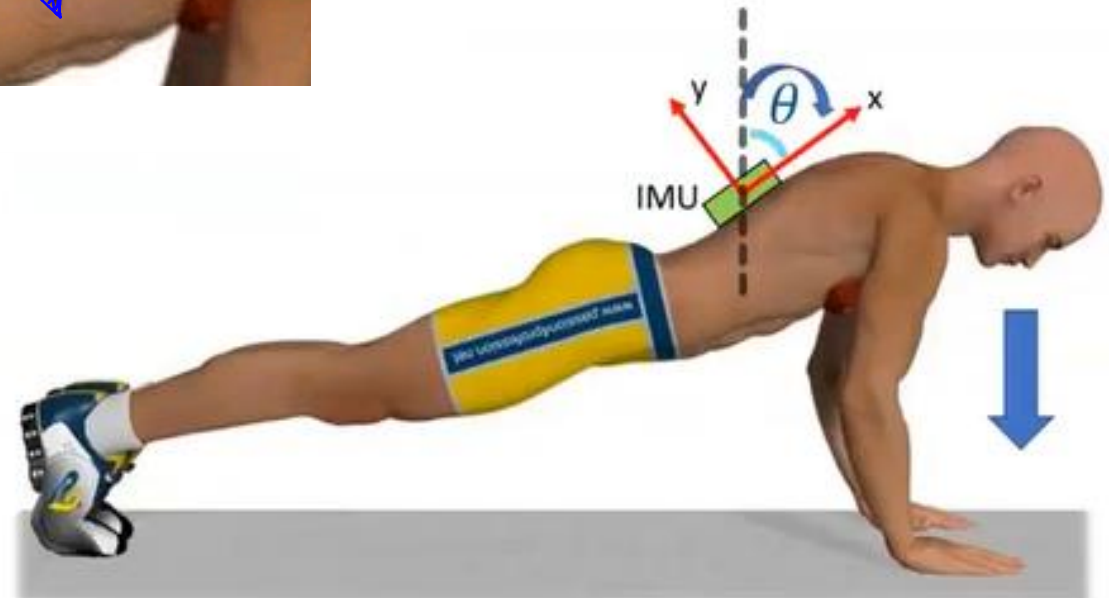
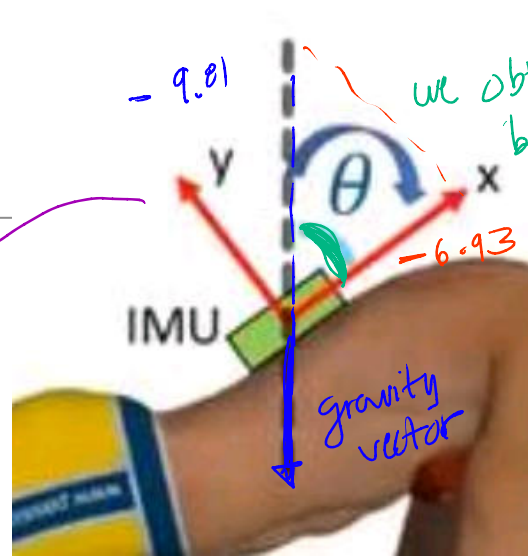
To find the components along x - y axis

$$IMU_y = -9.81 * \sin(\text{angle}_{\text{final}}) \frac{m}{s^2} = -8.5 m/s^2$$

we do with the final angle

$$IMU_x = x * \cos(\theta_{\text{final}})$$

$$IMU_y = y * \sin(\theta_{\text{final}})$$



EXERCISE 5

Let us consider another IMU setup, which is parallel to the ground. Magnetometer measurements acquired is:

IMU reference plane

$$\vec{s}^m = [40, 0, -30] \mu T$$

x y z

Compute inclination angle that belongs to the given measurement location.

if you remember, we have between the magnetic north and the magnetometer readings we have an inclination angle which is the one we want to find

IMU SENSOR



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SOLUTION

we assume the two systems are aligned

$${}^g\text{mag} = \begin{bmatrix} h \cos \alpha \\ 0 \\ -h \sin \alpha \end{bmatrix} \rightarrow {}^s y^m = \begin{bmatrix} h \cos \alpha \\ 0 \\ -h \sin \alpha \end{bmatrix}$$

$${}^s y^m_{[1]} = h \cos \alpha$$

$${}^s y^m_{[3]} = -h \sin \alpha$$

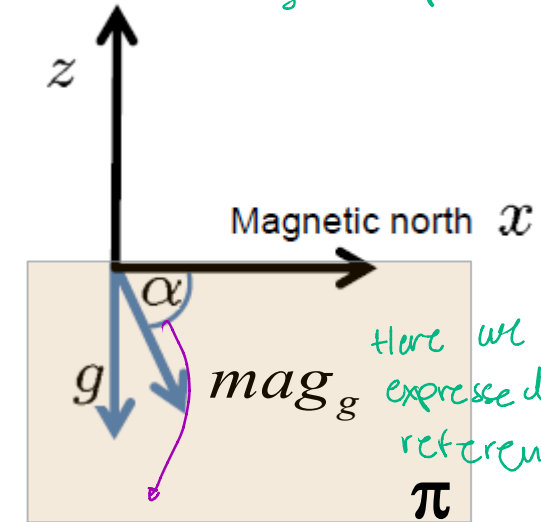
$$\rightarrow \frac{{}^s y^m_{[3]}}{{}^s y^m_{[1]}} = \frac{-h \sin \alpha}{h \cos \alpha}$$

cos we don't know h, we compute it with $\tan^{-1}(\alpha)$

$$\tan \alpha = \frac{{}^s y^m_{[3]}}{{}^s y^m_{[1]}} \rightarrow \alpha = \arctan \left[\frac{{}^s y^m_{[3]}}{{}^s y^m_{[1]}} \right] = 36.8^\circ$$

we simply obtain the angle

we can consider z-x cross section cos y component is 0



here we have it expressed in global reference frame.

alpha which is the angle between the magnetic north and the magnetometer

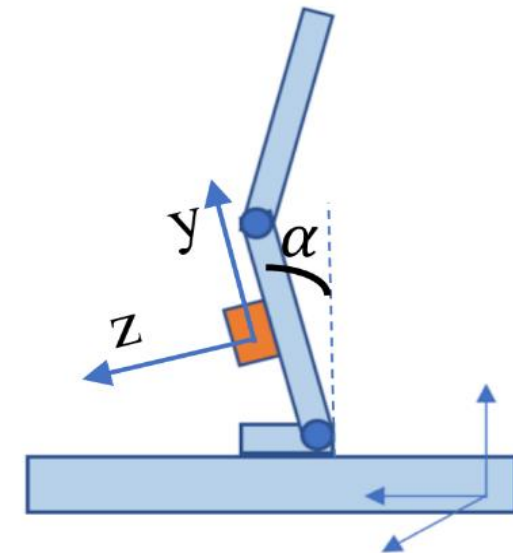
IMU SENSOR



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EXERCISE 6

An IMU is attached to the calf of the patient as seen in Figure. Calculate the value of α as the angle between the vertical axis and the longitudinal axis of the calf knowing that the output of the accelerometer is equal to $[0; -8.48; 4.9]m/s^2$



IMU SENSOR



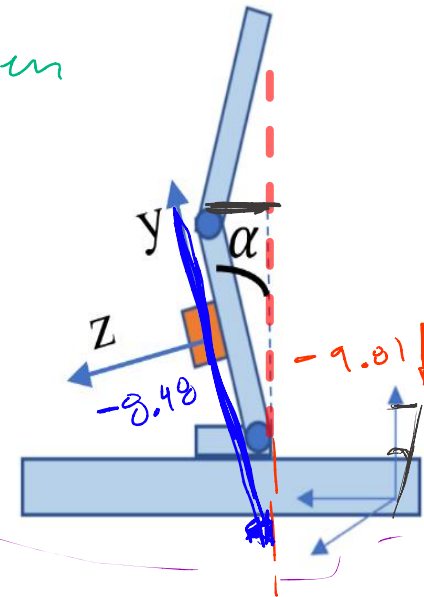
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SOLUTION

$$\alpha = \cos^{-1} \left(\frac{-8.48}{-9.81} \right) = 30.18^\circ$$

$[0, -8.48; 4.9]$ m/s²

we want to evaluate α
which is the angle between
y and the vertical
axis



$$\cos \alpha = \frac{y}{g}$$

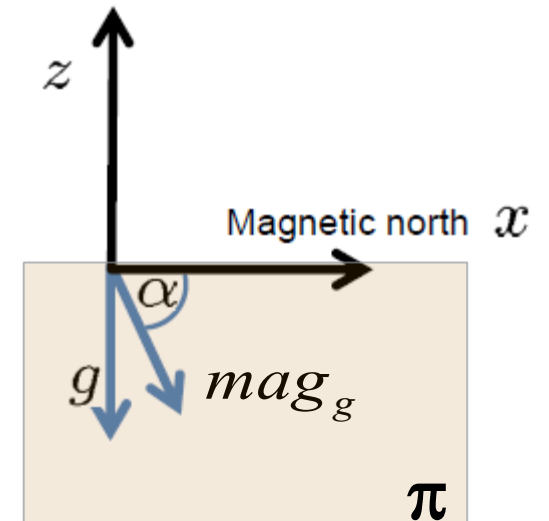
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EXERCISE 7

Estimate the magnetometer reading ${}^s y^m$ given that we are in Milan (magnetic field strength 47.4 mT and inclination angle 60°), the output of the accelerometer is constant over time and equal to $[0, 9.81, 0]m/s^2$. The body to which the IMU is attached has a $r_2 = [0 \ 0 \ 1]$.



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we have then magnetic field strength,
inclination angle, the output of the
accelerometer and vector r_2 of
the rotation matrix and we wanna
get S_{ym}

SOLUTION

${}^s y^m = {}^s R_g {}^g mag$

*rotation matrix x
mag vector
in the y ref system*

$${}^g mag = \begin{bmatrix} h \cos \alpha \\ 0 \\ -h \sin \alpha \end{bmatrix} = \begin{bmatrix} 0.0237 \\ 0 \\ -0.041 \end{bmatrix}$$

g_{mag} is easily found with h and α

$$h = 47.4 mT$$

$$\alpha = 60^\circ$$

$$r_2 = [0 \ 0 \ 1]$$

$${}^s y^a = [0; 9.81; 0]$$

${}^s R_g = [r_1; r_2; r_3]$ we move on with the computation of rotation matrix

$\frac{{}^s y^a}{\|{}^s y^a\|} = {}^s R_g \frac{{}^g g}{\|{}^g g\|}$

rotation matrix • gravity vector

$$\rightarrow {}^s y^a = [r_1 \ r_2 \ r_3] \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$\frac{{}^s y^a}{\|{}^s y^a\|} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\rightarrow {}^s y^a = -r_3$

*accelerometer data
is equal to $-r_3$*

$$\rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = -r_3 \rightarrow \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = r_3$$

then $r_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$

this is a
direction on
global frame

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SOLUTION

with $r_2 \times r_3$ we obtain r_1

$$r_1 = r_2 \times r_3 \rightarrow r_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$${}^s R_g = [r_1; r_2; r_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

we have now everything to estimate
readings in the step down

$${}^s y^m = {}^s R_g {}^g mag = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.0237 \\ 0 \\ -0.041 \end{bmatrix} = \begin{bmatrix} 0.0237 \\ -0.041 \\ 0 \end{bmatrix}$$

$$h = 47.4 mT$$

$$\alpha = 60^\circ$$

$$r_2 = [0 \ 0 \ 1]$$

$${}^s y^a = [0; 9.81; 0]$$

the magnetometer

EXERCISE 8

Consider setup of the IMU, with Z-Axis of IMU aligned with the normal vector of the ground. Compute the components of the magnetic field, supposing that your measures have been acquired in Paris (field inclination = 64° , field strength = 48205 nT). Consider magnetic north is aligned with the x-axis of the IMU and there is no component on the y-axis.

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SOLUTION

As before, we have inclination angle and magnitude of the field

$${}^g mag = \begin{bmatrix} h \cos \alpha \\ 0 \\ -h \sin \alpha \end{bmatrix}$$

magnetometer vector in global reference frame

$$\alpha = 64^\circ;$$

$$h = 48205 \text{ nT};$$

Also here we assume that the two reference systems are aligned. We don't compute rotational matrix

$${}^s y^m = \begin{bmatrix} h \cos \alpha \\ 0 \\ -h \sin \alpha \end{bmatrix} = \begin{bmatrix} 2.11 \times 10^{-5} \\ 0 \\ -4.33 \times 10^{-5} \end{bmatrix} m/s^2$$

The magnetometer data in the strap-down is simply

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EXERCISE 8

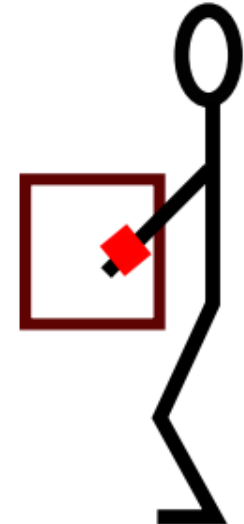
We are interested in measuring the acceleration values to which the subject's wrist in the picture is subjected during the lifting movements.

Consider that, when noise is negligible, the measured ^{linear}~~linear~~ acceleration s_y^a is:

$$s_y^a = [-4.900; 7.357; -4.247] \frac{m}{s^2}$$

The measured magnetic field is (*measurement are done in Milan):

$$s_y^m = [-0.399; 0.184; -0.174] * 10^{-4} T$$



REQUESTS

1.1 Estimate the linear accelerations to which the wrist is subjected, in the global reference frame.

1.2 Verify that the measurements have been done in Milan ($h = 47358nT$; $\alpha = 61$).

in this part, we have accelerometer readings in the strap-down and we wanna obtain the acceleration under which the wrist goes w.r.t the global reference frame.

$$\boxed{{}^s y^a} = {}^s \ddot{p} + {}^s g = {}^s R_g \left({}^g \ddot{p} + {}^g g \right)$$
$${}^g \ddot{p} = {}^g R_s {}^s y^a - {}^g g$$

We want the accelerometer in the global reference frame, then we compute the rotation matrix

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SOLUTION 1.1

$$\frac{s y^a}{\|s y^a\|} = {}^s R_g \frac{{}^G g}{\|{}^G g\|} \rightarrow s y^a = [r_1 \ r_2 \ r_3] \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \rightarrow \begin{bmatrix} -4.9 \\ 7.35 \\ -4.24 \end{bmatrix} = -r_3 \rightarrow \boxed{\begin{bmatrix} 4.9 \\ -7.35 \\ 4.24 \end{bmatrix}} = r_3$$

↘ gravity

$$\frac{s y^m \times s y^a}{\|s y^m \times s y^a\|} = r_2 \rightarrow \boxed{r_2 = \begin{bmatrix} 0.22 \\ -0.373 \\ -0.90 \end{bmatrix}}$$

then we obtain r_2 with the cross product of the magnetometer X accelerometer data.

$$r_1 = r_2 \times r_3 \rightarrow \boxed{r_1 = \begin{bmatrix} -0.83 \\ -0.546 \\ 0.021 \end{bmatrix}}$$

and $r_1 = r_2 \times r_3$

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SOLUTION 1.1

$\rightarrow {}^sR_g = \begin{bmatrix} -0.83 & 0.22 & 0.49 \\ -0.546 & -0.373 & -0.75 \\ 0.021 & -0.901 & 0.43 \end{bmatrix} \rightarrow {}^gR_s = \begin{bmatrix} -0.83 & -0.546 & 0.021 \\ 0.22 & -0.373 & -0.901 \\ 0.49 & -0.75 & 0.43 \end{bmatrix}$

or rotation matrix of step-down $\rightarrow$ *we want the rotation matrix for the global then*

${}^gR_s = {}^sR_g^{-1}$

${}^g\ddot{p} = {}^gR_s {}^s y^a - {}^g g$

Acceleration in the step-down

${}^g\ddot{p} = \begin{bmatrix} -0.83 & -0.546 & 0.021 \\ 0.22 & -0.373 & -0.901 \\ 0.49 & -0.75 & 0.43 \end{bmatrix} \begin{bmatrix} -4.9 \\ 7.35 \\ -4.24 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ -9.8 \end{bmatrix} \rightarrow {}^g\ddot{p} = \begin{bmatrix} 0.00 \\ 0.00 \\ 0.003 \end{bmatrix}$

gravity vector

The wrist is still!

it seems like
wrist is not
moving

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SOLUTION 1.2

Milan:

$$h = 47358 \text{ nT}$$

$$\alpha = 61^\circ$$

$$\rightarrow {}^g \text{mag} = \begin{bmatrix} h \cos \alpha \\ 0 \\ -h \sin \alpha \end{bmatrix} = \begin{bmatrix} 2.3 \\ 0 \\ -4.14 \end{bmatrix} * 10^{-5}$$

obtained with the magnitude and angle of the magnetometer reading

$${}^g \text{mag} = {}^g R_s {}^s y^m \rightarrow {}^g \text{mag} = \begin{bmatrix} -0.83 & -0.546 & 0.021 \\ 0.22 & -0.373 & -0.901 \\ 0.49 & -0.75 & 0.43 \end{bmatrix} \begin{bmatrix} -0.39 \\ 0.184 \\ -0.174 \end{bmatrix} * 10^{-4} = \begin{bmatrix} 2.3 \\ 0 \\ -4.14 \end{bmatrix} * 10^{-5}$$

Exactly the same

now
we compare it
with the values obtained with
the reading of the magnetometer the problem gave us and the
rotation matrix

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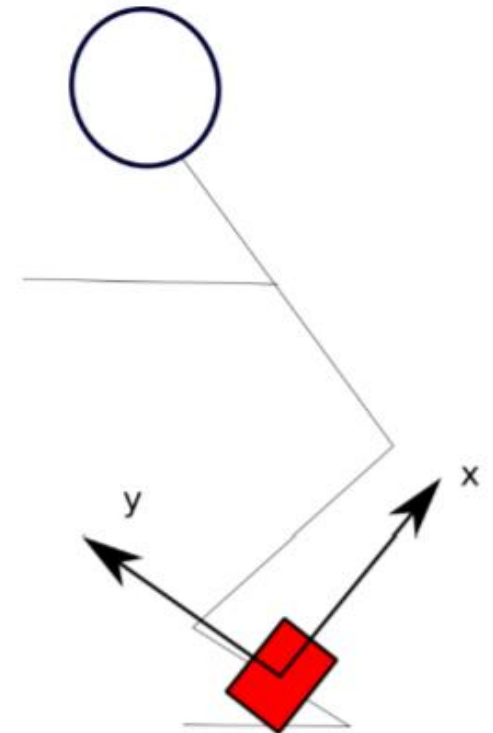
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EXERCISE 9

Right before jumping (body acceleration = 0), an IMU attached to the subject ankle (as in Figure) measures an acceleration along its y axis equal to -3.47 m/s^2 .

1.1 Compute the acceleration component measured along the IMU x axis.

1.2 Compute the components of the magnetic field, supposing that your measures have been acquired in Berlin (field inclination = 69° , field strength = 49822 nT). Consider Magnetic North aligned with x axis of the IMU and there is no components on the z axis



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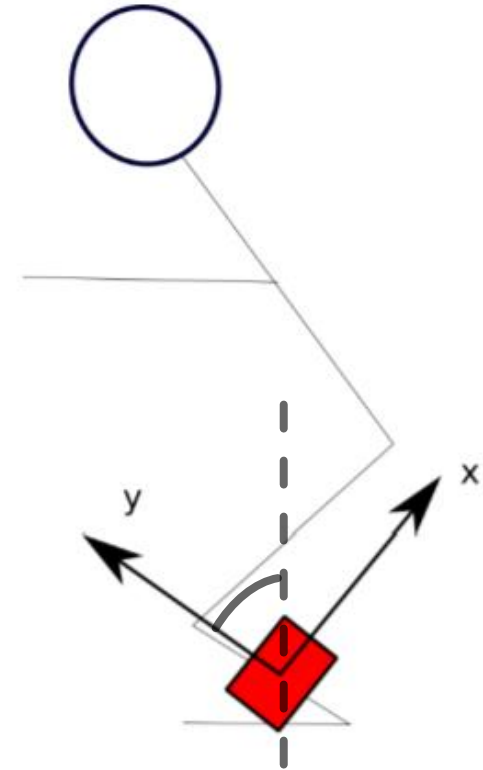
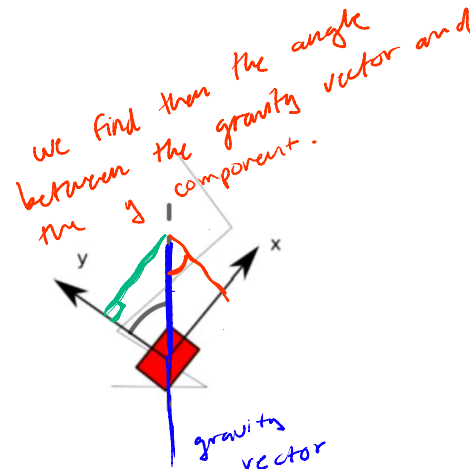
SOLUTION 1.1

we want the acceleration along the
X-axis of the IMU

$$\text{Angle: } \cos^{-1}\left(\frac{-3.47}{-9.81}\right) = 70^\circ$$

$$IMU_x = -9.81 * \sin(\text{angle}) \frac{m}{s^2} = -9.21 m/s^2$$

And then to find the IMU in the X-axis
with the $g \cdot \sin(\alpha)$



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*The second question asks
magnetometer readings*

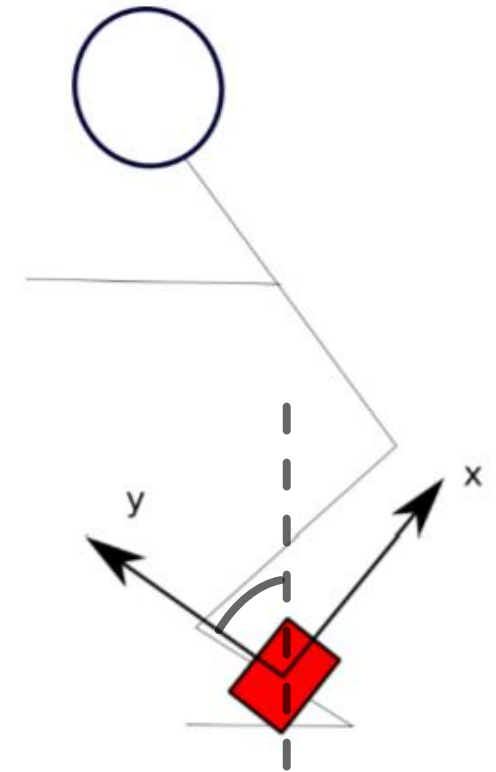
SOLUTION 1.2

$$\alpha = 69^\circ$$

$$h = 49822 \text{ nT}$$

*we know z-component is 0
and magnetic north (+) is
aligned with the x-axis*

$$^s_{mag} = \begin{bmatrix} h \cos \alpha \\ -h \sin \alpha \\ 0 \end{bmatrix} = \begin{bmatrix} 1.78 \times 10^{-5} \\ -4.65 \times 10^{-5} \\ 0 \end{bmatrix} T$$



MOVING AVERAGE FILTER



EXERCISE 11

During your experiments, you are required to design a digital moving average low-pass gravity detector filter $M = 3$ points. The signal is sampled at 150Hz.

Compute the filter 3dB cut-off frequency.

for the percentage of sampling
freq

$$F_c = \frac{0.443}{\sqrt{M^2 - 1}} \times 100$$

For three points we
have a narrow response.

$$f_{\text{sampling}} = 150 \text{ Hz}, \quad M = 3$$

then

$$f_{\text{cut-off}} = \frac{0.443}{\sqrt{M^2 - 1}} \times f_{\text{sampling}}$$

$$f_{\text{cut-off}} = 23.5 \text{ Hz}$$

MOVING AVERAGE FILTER



EXERCISE 12

In order to filter rectified EMG of a patient, a moving average FIR filter with 31 samples is used. What is its cut-off frequency (-3 dB) for a sampling rate of 500 Hz?

$$f_{\text{cut-off}} = \frac{0.443}{\sqrt{M^2 - 1}} \times f_{\text{sampling}}$$

$$f_{\text{cut-off}} = 7.14 \text{ Hz}$$

